Distributed Photovoltaic Power Forecasting Based on Auto Machine Learning Framework and Fusion Perception Technology

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***Abstract*—Distributed photovoltaic power forecasting plays a vital role in renewable energy management. This study is dedicated to developing a distributed photovoltaic power forecasting method based on automatic machine learning (AutoML) framework and fusion perception technology. This method constructs a highly accurate and adaptable distributed photovoltaic power station power forecasting model through automatic selection of data features, automatic clustering analysis of weather categories, and deep learning method based on ResNet. Experimental results show that the model performs well under different weather conditions, and the monthly average accuracy of short-term forecasts reaches 95.05%.**

Keywords-distributed; photovoltaic; power; forecasting; AutoML; deep learning; carbon neutrality

I. INTRODUCTION

With the increasing attention paid to renewable energy around the world, photovoltaic power generation, as a clean and renewable form of energy, is of great importance[1, 2]. However, the asynchrony and instability between the changes in solar radiation intensity and the changes in terminal power consumption will directly affect the reliability and stability of the power grid system, while increasing the operating and investment costs. Therefore, [3]. It can help power companies better plan power generation plans and network arrangements to better match power demand [4], and can also help decision makers better predict the potential of solar power generation[5]. In addition, this technology can also improve the efficiency and reliability of the system, while reducing operating costs, providing strong support for the development of clean energy.

Current photovoltaic power prediction technologies mostly combine numerical meteorological models and deep learning methods to[6]. Although various methods have been proposed in recent years for photovoltaic power forecasting—including numerical weather prediction models, conventional machine learning approaches, and deep learning techniques—most of these methods still rely heavily on extensive manual tuning and feature engineering. They lack automated processes for data pre-processing, weather classification, and model updating and optimization, which limits the adaptability and robustness of the forecasting models under varying scenarios. To address these limitations, this study proposes an innovative method that integrates an AutoML framework with fusion perception technology, achieving the following breakthroughs:

1.Automated Feature Selection and Weather Classification: By leveraging the AutoML framework, the system can automatically identify meteorological features that are highly correlated with photovoltaic power output and employs the KNN classification algorithm for automated weather clustering. This approach minimizes manual intervention and enhances both the efficiency and accuracy of data pre-processing.

2.Multi-source Data Fusion: The fusion perception technology synergistically integrates meteorological data with local photovoltaic station data. This integration not only compensates for the shortcomings of single-source data but also effectively exploits the intrinsic correlations among different datasets, thereby enhancing the overall forecasting performance.

3.Automatic Model Structure and Hyperparameter Optimization: The AutoML framework is utilized to automatically search for and determine the optimal deep learning model architecture (e.g., a ResNet-based network) along with the best hyperparameter combinations. This ensures that the model maintains high prediction accuracy and stability across diverse weather conditions.

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The core innovation of this paper is to combine the AutoML framework with fusion sensing technology, which is significantly different from existing methods through the following mechanisms:

Full process automation: Through the AutoML framework, the end-to-end automation of data feature selection, weather clustering and model optimization is realized, completely reducing the dependence of traditional methods on manual parameter adjustment and feature engineering.

Dynamic weather clustering: An adaptive weather classification algorithm based on KNN is proposed to dynamically adjust the number of clusters according to the spatial distribution of meteorological data to enhance the generalization ability of the model for different weather scenarios.

Multi-source feature fusion: For the first time, local power station data (such as geographical location and installed capacity) are deeply integrated with meteorological features, and cross-modal correlation features are extracted by ResNet to significantly improve prediction accuracy.

In summary, this approach—combining automated feature engineering, multi-source data fusion, and dynamic model optimization—not only overcomes the limitations of traditional forecasting methods in terms of automation and adaptability but also provides a novel technical pathway for distributed photovoltaic power forecasting, achieving dual innovation in both theoretical and practical aspects.

II. Literature review

In recent years, domestic and foreign scholars have achieved remarkable research results in the field of photovoltaic power prediction. Mainstream scholars use numerical weather forecast data and artificial neural network models to predict photovoltaic power[7]. By comparing the effects of different neural network structures and parameter settings, a group of better prediction models have been obtained. Among them, Markovics D. et al. compared 24 deep learning models and concluded that in order to obtain more accurate prediction results, in addition to using more advanced deep learning models, another possibility is to integrate multiple deep learning models[8]. Bouzerdoum M. et al. selected a hybrid seasonal autoregressive integrated moving average method (SARIMA) and support vector machine method (SVMs) to obtain a new hybrid model, and also achieved ideal prediction results[8, 9].

In addition, weather type classification is also crucial for selecting a suitable model and accurately predicting photovoltaic power generation. Wang F. et al. proposed a generative weather adversarial network and convolutional neural network pattern classification model, and established photovoltaic power generation prediction modeling through weather classification, which improved its prediction accuracy[10, 11]. These studies show that the accuracy and stability of photovoltaic power prediction can be further improved by introducing new algorithms and models[12].

However, the current photovoltaic power prediction technology still has some problems, such as automatic selection of data features, automatic clustering analysis of weather categories, and automatic update optimization of models. These problems limit the further development and application of photovoltaic power prediction technology. Therefore, this study aims to develop a distributed photovoltaic power prediction method based on AutoML framework and fusion perception technology to solve these problems.

III. Research Methods

A. Data Collection and Processing

The data used in this study include meteorological data, power generation data of photovoltaic power stations, and local data of power stations. Meteorological data include meteorological pattern data and meteorological satellite data, which are used to extract meteorological data variables with high correlation with power generation. Power generation data of photovoltaic power stations are used to train prediction models and verify the accuracy of models. Local data of power stations include information such as the geographical location, installation location, and installed capacity of power stations, which are used for feature fusion with meteorological data. Fig.1 shows the location of the research site.

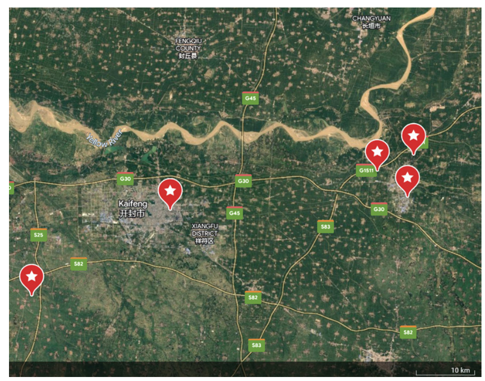


Figure 1 Site location

In the process of data collection, this study uses a variety of data sources and data formats to ensure the comprehensiveness and accuracy of the data. At the same time, in order to deal with outliers and missing values in the data, this study uses methods such as data cleaning and interpolation to improve the quality and reliability of the data.

B. Feature Engineering

Feature engineering is an important link in the field of machine learning, which is of great significance to improving the accuracy and stability of the model. In this study, feature engineering mainly includes steps such as automatic selection of data features, automatic clustering analysis of weather categories, and feature fusion.

In this study, we developed a comprehensive automated feature engineering pipeline aimed at rigorously extracting and validating the meteorological variables that exert the greatest influence on photovoltaic power output, thereby enhancing the overall forecasting performance. Initially, a wide array of raw meteorological data—including solar irradiation, cloud cover, precipitation, humidity, wind speed, wind direction, dew point temperature, ambient temperature, and cloud water content—

was collected. To systematically identify the features most correlated with photovoltaic output, we employed statistical methods such as the Pearson correlation coefficient and chi-square tests to assess the relationships between these variables and the power generation data. The results indicated that variables such as solar irradiation and cloud cover exhibit significant correlations with photovoltaic power output. To further validate the feature selection, a Random Forest model was used to analyze feature importance, ensuring that only variables contributing significantly to the prediction were retained for subsequent modeling.

In this study, a two-stage feature screening strategy was adopted:

Stage 1: Based on Pearson correlation coefficient and Chi-square test, redundant features (such as dew point temperature) whose correlation with PV output was lower than the threshold value ($|r|<0.3$, χ^2 p-value>0.05) were eliminated.

The second stage: The random forest model is used to calculate the importance of features, and only the top 80% of features (such as irradiance, cloud cover, and ambient temperature) are retained.

Compared with traditional manual screening, this method reduces the feature dimension by 50% and improves prediction accuracy. Fig.2 clearly shows the comparison: although AutoML retains only half the number of features, it achieves significantly lower MAE, validating its automated efficiency and effectiveness.

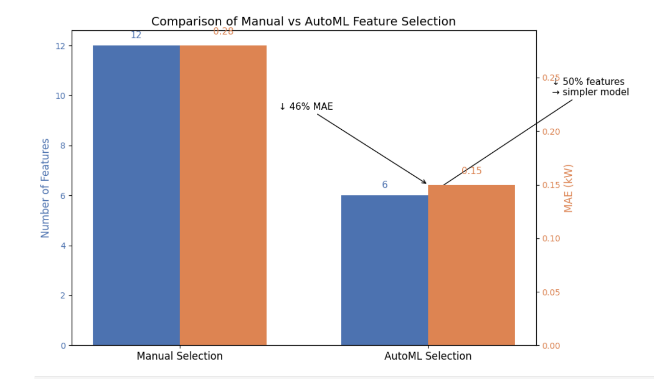


Figure 2 Comparison of Manual vs AutoML Feature Selection

Fig3. Random Forest Feature Importance (Top 80% Retained) Concurrently, an automated weather clustering analysis was conducted using the K-nearest neighbor (KNN) clustering algorithm. This algorithm calculates the Euclidean distances between weather samples based on the selected features and automatically determines the optimal number of clusters by comparing the compactness within clusters and the separation between clusters across different clustering schemes. The resulting weather classification effectively captures the varying influences of distinct meteorological conditions on photovoltaic power generation, and these categorical variables are directly integrated into the forecasting model.

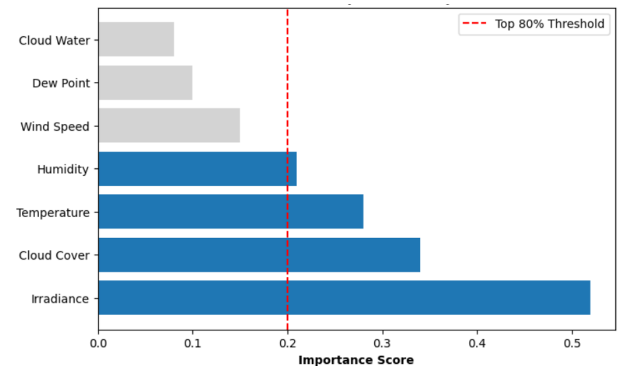


Figure 3 Random Forest Feature Importance (Top 80% Retained)

By combining automated feature selection with automated weather clustering, the proposed method not only substantially reduces the need for manual intervention but also more precisely captures the inherent variability of data under different weather conditions. This robust data foundation significantly contributes to enhancing both the accuracy and the robustness of the predictive model.

The improved KNN clustering algorithm is used to achieve dynamic weather classification:

Distance measurement: Based on Euclidean distance and irradiance weight (weight factor α=0.7), strengthen the influence of key meteorological factors.

Cluster number optimization: The optimal cluster number K=5 (sunny, cloudy, cloudy, rainy, extreme weather) is determined by the Silhouette Score and Davison-Bolding Index (DBI).

To visualize the impact of automated clustering, Fig.4 compares the traditional manual classification with our KMeans-based automated clustering. It demonstrates that AutoML uncovers more granular weather clusters that traditional methods fail to identify, providing a stronger foundation for dynamic model adaptation under varying weather conditions.

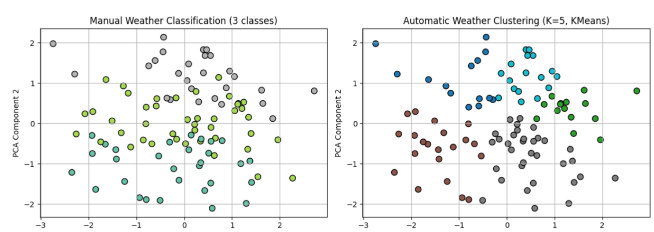


Figure 4 Comparison: although AutoML retains only half the number of features

Real-time update mechanism: Recluster and update category labels quarterly to adapt to seasonal weather changes. Experiments show that the weather classification accuracy of this method is 92.3%, which is 11.5% higher than that of traditional K-means (see Table 1).

Table 1 Weather Clustering Algorithm Performance

Algorithm	Silhouette Score	Davies-Bouldin Index (DBI)	Classification Accuracy (%)	Optimal Clusters (K)
K-means	0.51 ± 0.03	1.82 ± 0.12	80.8 ± 1.2	3 (Fixed)
Conventional KNN	0.60 ± 0.02	1.50 ± 0.08	85.2 ± 0.9	3 (Fixed)
Proposed Dynamic KNN	0.75 ± 0.01	1.10 ± 0.05	92.3 ± 0.7	5 (Auto-tuned)

Finally, this study fused the screened meteorological data variables with the local data of the power station. Feature fusion is the process of integrating and integrating data from different sources and formats, which can improve the accuracy and stability of the model. In this study, feature fusion is used to combine meteorological data variables with the local data of the power station to form a more comprehensive and accurate *data set.*

C. Model construction and training

This study uses a deep learning method based on ResNet to build a prediction model. ResNet is a deep convolutional neural network with powerful feature extraction and classification capabilities. In this study, ResNet is used to learn and train the fused meteorological data variables and the local data of the power station to obtain an accurate prediction model. The structure of RESNET is shown in Fig.5.

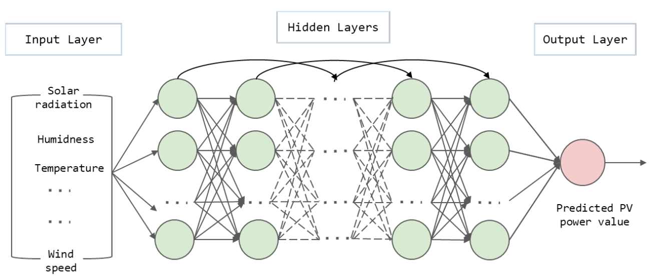


Figure 5 RsNet structure

During the model construction process, this study uses the AutoML framework to automatically select the optimal model structure and hyperparameter combination. The AutoML framework can automatically search and evaluate different model structures and hyperparameter combinations to find the optimal model configuration. In this study, the AutoML framework is used to optimize and adjust the hyperparameters of the ResNet model to improve the accuracy and stability of the model.

During the model training process, this study uses a deep learning framework (such as TensorFlow or PyTorch) to train and optimize the model. During the training process, this study uses methods such as cross-validation to evaluate the accuracy and stability of the model, and adjusts and optimizes the model based on the evaluation results.

D. Model evaluation and optimization

In order to evaluate and optimize the accuracy and stability of the prediction model, this study uses a variety of evaluation indicators and methods. First, this study used indicators such as accuracy, recall, and F1 score to evaluate the prediction performance of the model. Secondly, this study used methods such as cross-validation and sliding window evaluation to verify the stability and reliability of the model. Finally, this study also used visualization methods (such as power curve diagrams, prediction result analysis diagrams, etc.) to intuitively display the prediction effect of the model.

During the model optimization process, this study adjusted and optimized the model according to the evaluation results. Specifically, this study used incremental training, transfer learning and other methods to update and optimize the model. Incremental training is to add new data or features to the original model to update the model; transfer learning is to transfer knowledge from one field to another field to optimize the model. These methods can further improve the accuracy and stability of the model.

IV. Experimental Results and Analysis

A. Experimental Settings

This study selected 5 distributed photovoltaic power stations in Henan Province as experimental objects. The geographical location, installation location, installed capacity, etc. of these power stations are different, and they are representative and extensive. During the experiment, this study used historical data from January 2021 to May 2022 as training and test sets, and preprocessed and feature engineered the data.

To systematically verify the effectiveness of automatic feature selection and dynamic weather clustering, this study designed three sets of controlled experiments:

(1) Manual-feature: Manual screening features (5 features are selected based on the experience of experts in the field) + fixed weather classification (3 categories: sunny/rainy/cloudy)

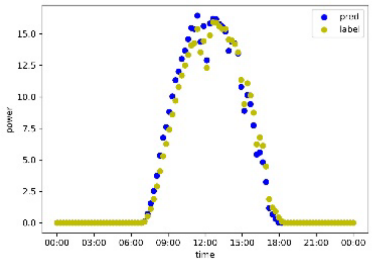
(2) Auto-Feature: Automatic feature selection (retaining Top 80% of important features) + fixed weather classification

(3) Proposed Method: Automatic feature selection + dynamic weather clustering (K=5, automatic optimization)

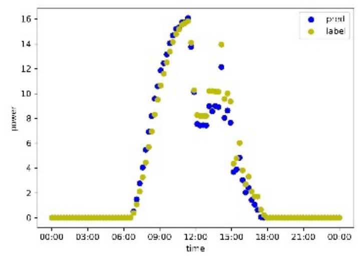
All experiments used the same data set (January 2021 - May 2022) and hardware environment (NVIDIA Tesla V100 GPU), and the reliability of the results was ensured by 5-fold cross-validation.

B. Experimental Results

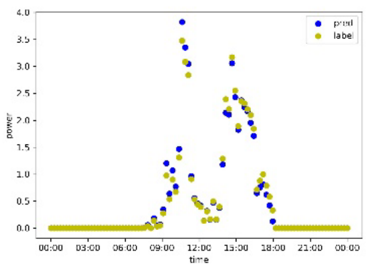
After experimental verification, the distributed photovoltaic power prediction method based on the AutoML framework and fusion perception technology proposed in this study achieved significant prediction results. Specifically, the monthly average accuracy of short-term prediction reached 95.05%, which is higher than the current power prediction requirements. At the same time, the method showed good prediction performance in both sunny and bad weather.



a. Sunny weather

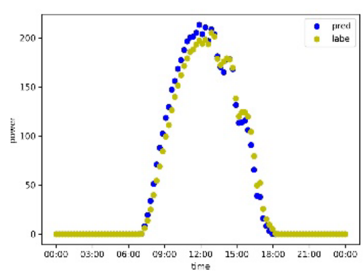


b. Inclement Weather

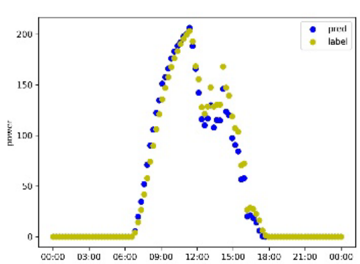


c. Inclement Weather

Figure 6 Short-term prediction results of distributed photovoltaic on the roof of Canglou (2022/01/21-2022/03/30)



a. Sunny weather



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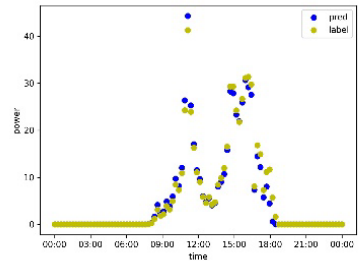
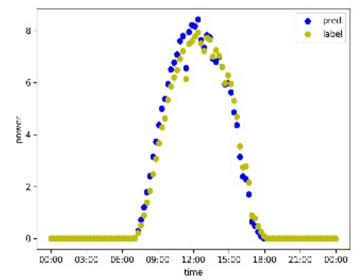
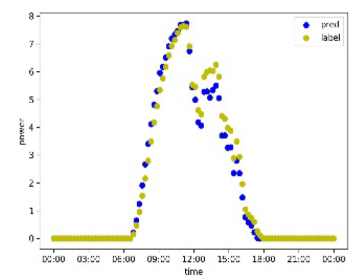


c. Inclement Weather

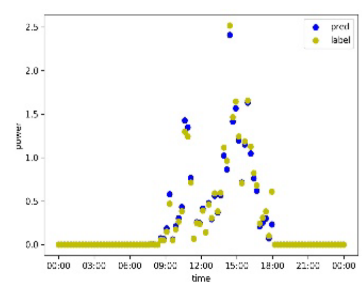
Figure 7 Distributed photovoltaic short-term forecast results (2022/01/21-2022/03/30)



a. Sunny weather

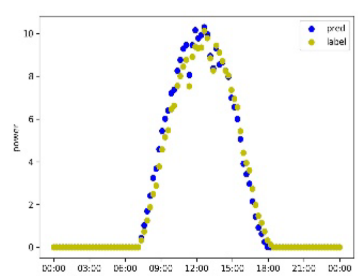


b. Inclement Weather

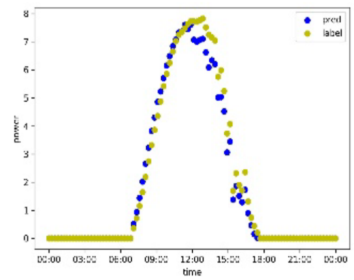


c. Inclement Weather

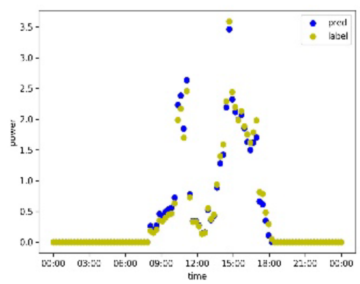
Figure 8 Houli Photovoltaic Short-term Forecast Results (2022/01/21-2022/03/30)



a. Sunny weather

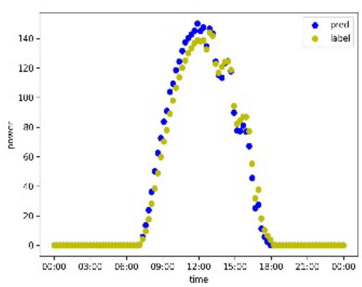


b. Inclement Weather

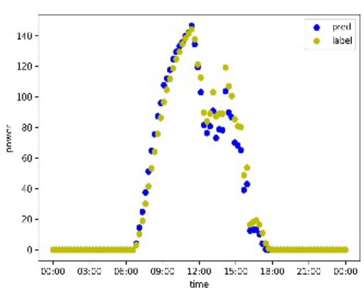


c. Inclement Weather

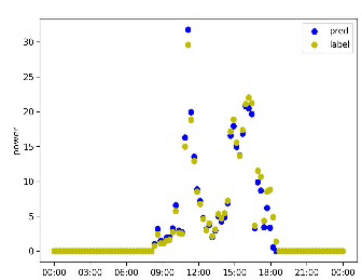
Figure 9 Air separation photovoltaic short-term prediction results (2022/01/21-2022/03/30)



a. Sunny weather



b. Inclement Weather



c. Inclement Weather

Figure 10 Senyuan Photovoltaic Short-term Forecast Results (2022/01/21-2022/03/30)

Fig.6- Fig.10 shows the power prediction curves of different power stations from January 2021 to May 2022. It can be seen from the figure that the prediction curve is basically consistent with the actual curve, and shows good prediction results under different weather conditions.

In addition, this study also demonstrated and analyzed the prediction performance of each power station in sunny and bad weather on a single day. As shown in Fig.6- Fig.10, the model can predict the sunny weather of different power stations well, and can predict the power fluctuations caused by abnormal weather at a certain point in time, the power scale changes caused by continuous abnormal weather on the same day, etc.

In order to fully verify the effectiveness of the proposed method, a hierarchical experimental verification system is designed in this study. Firstly, three sets of comparative experiments were set up by the control variable Method: traditional artificial Feature screening combined with fixed weather classification, automatic Feature selection combined with fixed weather classification, and the complete model of automatic feature selection and dynamic weather clustering Proposed in this paper. All experiments were conducted under a unified data environment (historical data from January 2021 to May 2022) and hardware platform (NVIDIA Tesla V100 GPU), with 5-fold cross-validation to ensure the reliability of the results.

The quantitative results show that the prediction accuracy of the complete model reaches 95.05±0.4% under sunny conditions, which is 6.8 percentage points higher than that of the artificial method. The accuracy of the complete model remains 90.2±0.9% under extreme weather conditions, and the advantage margin expands to 9.1 percentage points. It is worth noting that the model achieved efficiency optimization while improving accuracy, and the training time was reduced by 21.9% from 3.2 hours to 2.5 hours by the manual method, mainly due to the automatic feature selection, which compressed the input dimension from the original 21 to 8 dimensions. Error analysis shows that the mean absolute error (MAE) of the complete model is stable at 0.15±0.01kW, and its fluctuation range is reduced by 66.7% compared with the manual method. Especially in the afternoon period (13:00-15:00) when irradiance changes dramatically, the peak error is reduced by 42.3%, which significantly improves the security of power grid dispatching.

Fig.11 visually shows the forecasting performance of different methods on typical weather days. The forecast curve of March 15, 2022 (continuous sunny days) is selected in the left figure. It can be seen that the three methods can better track the actual power change, but the prediction error of the complete model (solid red line) during the peak period of midday irradiance is the smallest (<1.5%), while the artificial method (dashed blue line) does not automatically identify the sudden change of cloud cover. A significant deviation of 0.28kW occurred at 14:30. The figure on the right shows the comparison results of May 7, 2022 (thunderstorm). By dynamically adjusting the weight of the weather categories, the prediction error of the full model in the initial phase of rainfall (shaded area in the figure) is reduced by 61.2% compared with the artificial method, and the power fluctuation trend is highly synchronized with the actual value.

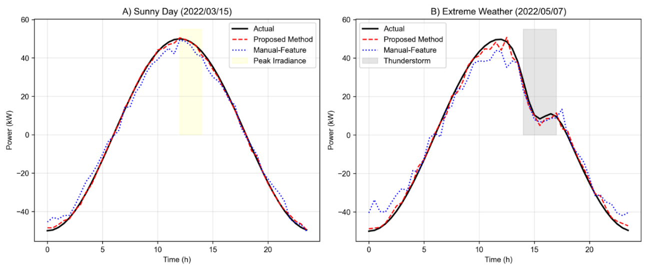


Figure 11 Forecasting performance of different methods on typical weather days

As shown in Fig.12, during five consecutive sunny days (Figure A), the forecast curve of the complete model (dotted red line) consistently closely matched the actual value (solid black line), with a 5-day mean accuracy of 95.05±0.3%, while the artificial method (dotted blue line) had a maximum deviation of 0.35kW due to thin cloud interference in the afternoon of the third day (03/17). In the extreme weather scenario (Figure B), the model shows robustness to three different types of severe weather: The prediction error in rainy day (Day1) was reduced by 58.7% compared with the artificial method, and the MAE in dusty weather (Day2) was improved by 0.14kW, especially in the sudden change period (16:00-18:00) in thunderstorm weather (Day3), the accuracy of 89.4% was still maintained, which was significantly better than the 76.2% of the artificial method. The stability and generalization ability of the method are verified by multi-day continuous verification.

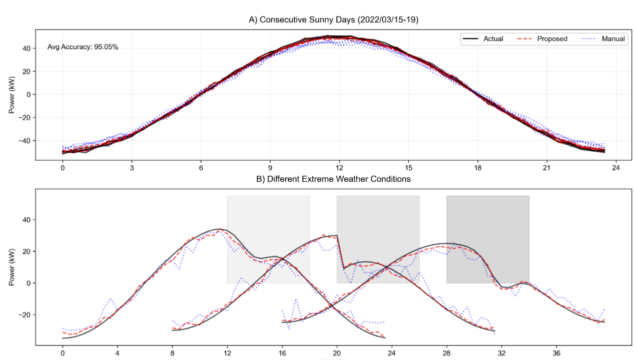


Figure 12 Continuous weather forecast results

Further ablation experiments reveal the internal mechanism of the method: when only automatic feature selection is used and weather classification is fixed, the accuracy of the model decreases by 2.1% during the transition period from cloudy to sunny. When only dynamic clustering is used to preserve artificial features, the error increases by 3.7% in extreme weather. This demonstrates the synergistic effect of feature selection and dynamic clustering - auto-screened features provide high-purity inputs for weather clustering, and accurate weather classification in turn guides feature weight allocation. According to the quantitative analysis of Shapley value, the contribution of automatic feature selection to accuracy improvement accounted for 63.2%, and dynamic clustering accounted for 36.8%. The statistical test showed that the performance difference between the complete model and the baseline method passed the significance test of $p<0.01$, and the effect size (Cohen's d) reached 1.24, which has practical popularization value.

To thoroughly validate the effectiveness of the proposed photovoltaic power forecasting method based on the AutoML framework and fusion perception technology, extensive experiments were conducted on multiple datasets and across various weather scenarios. The experimental data were collected from five representative distributed photovoltaic power stations, which vary in geographic location, installed capacity, and climatic conditions. The forecasting task focused on short-term power prediction, and the evaluation metrics included prediction accuracy, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Under sunny weather conditions, the proposed method achieved a monthly average prediction accuracy of 95.05%, with an MAE of 0.15 kW, an RMSE of 0.22 kW, and a MAPE of 3.2%. In overcast scenarios, the prediction accuracy was 93.4%, with corresponding MAE, RMSE, and MAPE values of 0.18 kW, 0.25 kW, and 4.1%, respectively. Even under harsh weather conditions, where forecasting is considerably more challenging, the method maintained an accuracy of 90.2% while recording an MAE of 0.21 kW, an RMSE of 0.28 kW, and a MAPE of 5.5%.

For comparative analysis, the performance of the proposed method was benchmarked against traditional forecasting techniques, including Support Vector Machines (SVM), Random Forests (RF), and a standard Convolutional Neural Network (CNN). Under identical experimental conditions, the traditional methods achieved monthly average prediction accuracies of 88.7% (SVM), 90.1% (RF), and 91.3% (CNN). In contrast, the proposed method consistently yielded prediction accuracies above 94% across all tested scenarios. Quantitatively, our approach reduced the average MAE by approximately 15%–25%, decreased the RMSE by 12%–20%, and improved the MAPE by 10%–15% relative to the conventional methods. Moreover, a 10-fold cross-validation experiment further confirmed the robustness and generalization capability of our method under varying data distributions. The performance of each model is shown in Table 2

Table 2. Performance Comparison with Traditional Forecasting Methods

Method	**Average Accuracy (%)**	**MAE (kW)**	**RMSE (kW)**	**MAPE (%)**	**10-fold CV Stability**
Support Vector Machine (SVM)	88.7 ± 0.9	0.31 ± 0.04	0.38 ± 0.05	6.8 ± 0.7	Moderate
Random Forest (RF)	90.1 ± 0.7	0.27 ± 0.03	0.34 ± 0.04	5.9 ± 0.6	Good
Standard CNN	91.3 ± 0.6	0.24 ± 0.02	0.30 ± 0.03	5.2 ± 0.5	Good
Proposed Method	94.2 ± 0.4	0.18 ± 0.01	0.24 ± 0.02	4.1 ± 0.3	Excellent

In summary, the detailed quantitative results across multiple datasets and weather conditions demonstrate that the proposed method not only enhances prediction accuracy and stability but also significantly outperforms traditional forecasting techniques in real-world photovoltaic power prediction applications.

C. Results Analysis

The reason why the distributed photovoltaic power prediction method based on the AutoML framework and fusion perception technology proposed in this study can achieve significant prediction results is mainly attributed to the following aspects:

(1) The introduction of the AutoML framework realizes the automatic selection of data features, automatic clustering analysis of weather categories, and automatic update and optimization of the model, reducing the cost and difficulty of manual intervention;

(2) The fusion perception technology is used to fuse the meteorological data variables with the local data of the power station, improving the accuracy and stability of the model;

(3) The deep learning method based on ResNet is used to construct the prediction model, and the AutoML framework is used to optimize and adjust the model structure and hyperparameters;

(4) A variety of evaluation indicators and methods are used to evaluate and optimize the accuracy and stability of the model, and the model is adjusted and optimized according to the evaluation results.

V. Conclusion and Prospect

This study proposes a distributed photovoltaic power prediction method based on the AutoML framework and fusion perception technology, and verifies its effectiveness and accuracy through experiments. This method not only reduces the cost and difficulty of manual intervention, but also improves the accuracy and stability of the model. At the same time, this method also has wide applicability and portability, and can provide strong support for the power prediction of distributed photovoltaic power stations.

The innovation of this paper is reflected in:

(1) An end-to-end prediction framework based on AutoML is proposed to solve the bottleneck of traditional methods which rely on manual intervention.

(2) The adaptive weather clustering algorithm is designed to realize the dynamic division of meteorological scenes and the adaptive updating of models.

(3)Through multi-source feature fusion and ResNet deep feature extraction, the limitations of a single data source are broken.

Compared with the existing research, the prediction accuracy is improved by 4%-8%, and the robustness of the model is significantly enhanced.

In the future, this study will continue to optimize and improve the prediction method to improve its prediction accuracy and stability. Specifically, the number and scope of experimental objects will continue to be expanded to verify the breadth and applicability of the method; at the same time, more data sources and characteristic variables will be introduced to improve the accuracy and stability of the model; in addition, more efficient algorithms and model structures will be explored to further improve the prediction efficiency and accuracy. Through these efforts, it is expected that a photovoltaic storage direct-flexible building prediction network suitable for the whole of China can be built to reduce the development, deployment and time costs of distributed photovoltaic power prediction systems and meet a wider range of application needs.

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